

Lesson 131 (4.0) Ground

Lesson Objective

The purpose of this lesson is to introduce VFR cross-country planning, weather, airspace, and regulations applicable to cross-country flight.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- During this lesson quiz the student on the following and provide instruction where needed.
- Student questions.

References

- Aviation Weather Services Handbook ([FAA-H-8083-28A](#))
- Aeronautical Information Manual. ([AIM](#))
 - Chapter 03 - Airspace.
 - Chapter 04 - Air Traffic Control.
 - Section 03 - Airport Operations.
 - Chapter 06 - Emergency Procedures.
 - Chapter 07 - Safety of Flight.
- Federal Aviation Regulations.
 - Suggested study list (Private/Recreational).
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 07 - Aircraft Systems.
 - Chapter 08 - Flight Instruments.
 - Chapter 12 - Weather Theory.
 - Chapter 13 - Aviation Weather Services.
 - Chapter 15 - Airspace.
 - Chapter 16 - Navigation.
- Pilot's Information Manual.
 - Section 01 - General.
 - Section 07 – Systems
- [FAA Order Jo 7110.118B](#)

Instructor Guidance

- When covering weather products, focus on how each product is used in real flight planning, not just definitions. Encourage students to explain how METARs, TAFs, winds aloft, and radar imagery influence their go/no-go decision.
- When repeating this lesson, quiz the student on previously completed line items to ensure continued proficiency.
- When teaching airspace, use a sectional chart and trace an actual cross-country route so the student can identify the airspace classes they will encounter along the route.

- When discussing diversions, stress that diverting early is usually the safest decision and that Pilots should not hesitate to change plans if conditions deteriorate.
- Airspace concepts are revisited in this lesson at a more advanced level.

Lesson Content

Weather Information

- Weather briefings:
 - Briefing Sources:
 - 1-800-WX-BRIEF (Leidos Flight Service)
 - 1800wxbrief.com
 - Approved EFB apps (ForeFlight, Garmin Pilot, etc.)
 - aviationweather.gov is an official FAA weather source
 - But it does not log the briefing like 1800wxbrief.
 - Types of weather briefings:
 - Outlook.
 - Standard.
 - Abbreviated.
 - In-flight.
 - For the purpose of this lesson, show the student how to obtain a weather briefing by calling the briefer or online.
 - Telephone:
 - Call 1-800-992-7433 → say "briefer" → say "Southern Florida".
 - Give the briefer the flight plan information (use the reference below if needed) and request a standard weather briefing.
 - Have the weather products opened on aviationweather.gov during the briefing to aid student understanding.
 - Online:
 - Navigate to 1800wxbrief.com and have the student create an account.
 - Proceed to the plan & brief tab and input all of the required flight plan information.
 - information (use the reference on page 4 if needed).
 - Click on route brief and select standard briefing.
- METAR:
 - Quiz the students on metar valid times.
 - Pull up sample METARs with less common reports on them to see if the student knows or can figure out what is being observed.
- TAF:
 - Quiz the student on TAF valid times.
 - Pull up sample TAFs with less common reports on them to see if the student knows or

can figure out what is being forecast.

- GFA Tool:
 - Show the student where to find the GFA tool.
 - Replaced older products such as Area Forecast & Low-level Significant charts
 - Go over the different items that the GFA tool can display:
 - Visibility, ceilings, cloud coverage, METARs, TAFs, precipitation, winds, thunderstorms, low level wind shear, turbulence, and icing.
- Winds and temperatures aloft chart:
 - A summary of winds aloft forecast for the route/area as interpolated from forecast data for the local and/or the adjacent reporting locations for level through 12,000 feet.
 - Winds and temps at specific locations.
 - Issued 4 times a day.
 - Wind is in true knots.
 - Temperature in Celsius.
 - No winds are forecasted within 1,500 AGL of station elevation.
 - No temperatures are forecasted within 2,500 AGL of station elevation.
 - If wind is forecasted to be 100-199 knots, subtract 50 from the wind direction and add 100 to the speed.
 - 731960
 - Wind is heading 230 at 119 knots.
 - The Temperature is -60 C.
 - If wind is at least 200 knots, wind will be coded as 99 knots.
 - 7799.
 - Wind is 270 and greater than 199 knots.
 - Calm or less than 5 knots is 9900.
 - Most weather reporting applications decode this for you.
- Radar Imagery:
 - National radar:
 - Shows the intensity of precipitation.
 - Provides a single view of the entire country.
 - Modern [NEXRAD](#) every 4 to 6 minutes in precipitation mode and every 10 minutes in clear air mode. Cockpit-displayed NEXRAD via ADS-B generally updates every 5 minutes (regional) or 15 minutes (national), but data can be 12-20+ minutes old due to processing delays.
 - NEXRAD (WSR-88D): Updates every 4–6 minutes during precipitation.
 - ADS-B Radar: Regional updates every 5 minutes; National updates every 15 minutes.
 - ForeFlight Radar: Updates every 5 minutes, but images are often 7–11 minutes old.
 - Loops the latest 2 hours of images.

- Single site radar:
 - Composed of 140 radar sites across the U.S.
 - Each site shows base reflectivity, composite reflectivity, and total precipitation.
 - Updated every 10 minutes.
 - Loops the latest 2 hours of images.
- Surface analysis chart:
 - Surface analysis charts – Chart depicts the distribution of several items including sea level pressure, the position of the highs, lows, ridges, and troughs, the location and type of fronts, wind direction and speed, temperature and dew point.
 - Analysis of current weather.
 - Computerized.
 - Transmitted every 3 hours.
 - Valid time shown on chart (observed data).
 - Found on [Weather Prediction Center](#)
 - High- and low-pressure systems:
 - In the northern hemisphere – flow of air from high to low pressure is deflected to the right due to the Coriolis force.
 - High pressure systems (outward - downward - clockwise).
 - Dry, stable, descending air.
 - Good weather.
 - Low pressure systems (inward - upward - counterclockwise).
 - Air flows in a low-pressure area to replace rising air.
 - Bad weather.
- Fronts:
 - Air masses (body of air with equal or uniform properties (humidity, temperature, pressure etc.)) form over regions and move the air over different areas which become subjected to varying conditions of the land or water which modify the nature of the air mass.
 - Warm front:
 - Occurs when the warm mass of air advances and replaces a body of colder air. Warm fronts move slowly, 10-25 mph. Slope of the advancing front slides over the top of the cooler air and gradually pushes it out of the area. As the warm air is lifted, the temperature drops, and condensation occurs from the humidity of the air.
 - Generally, cirriform, stratiform clouds (stable air) along with fog can be expected to form along the frontal boundary. (rain evaporates when it gets close to the cooler ground).
 - Light to moderate steady precipitation.
 - Flight toward a warm front:
 - Visibility decreases.
 - Falling barometric pressure.
 - Weather overall deteriorates.

- May be impossible to continue a VFR flight after flying through a warm front.
- Cold front:
 - Occurs when a mass of cold, dense and stable air advances and replaces a body of warmer air.
 - Moves at 25-30 mph, can move as fast as 60 mph.
 - Rapidly ascending air causes the temperature to decrease suddenly, forcing creation of clouds.
 - It "lifts moist air and creates instability" - recipe for thunderstorms.
 - Prior to cold front passing, cirriform towering cumulus clouds are present, and cumulonimbus clouds may develop.
 - Showery precipitation.
 - After frontal passage, clouds dissipate to cumulus clouds with a corresponding decrease in a precipitation.
 - Good visibility prevails and winds from the west and northwest.
 - Flight toward a cold front:
 - Flight progresses to the oncoming cold front; the clouds show signs of vertical development with a broken layer.
 - Pilots should use sound judgment on frontal passage.
- Stationary Front:
 - When the forces of two air masses are relatively equal, the boundary layer or front that separates them remains stationary and influences the local weather for days.
- Occluded Front:
 - Fast moving cold front catches up with a slow-moving warm front. Warm front weather prevails but is followed by cold front weather.
- Pilot reports (PIREPs) ([AIM](#))
 - Weather report from a flight deck. Consists of message type, location, time, altitude/flight level, type aircraft, and at least one other element to describe the report phenomena.
 - 2 types: routine (UA) and urgent (UUA):
 - Urgent includes:
 - Tornadoes.
 - Severe or extreme turbulence.
 - Hail.
 - Low level wind shear within 2000 ft.
 - Contains information regarding conditions as they actually exist.
 - Pilots confirm height of bases of clouds, locations of wind shear and turbulence, and icing.
 - Pilots are always encouraged to contact ATC or FSS with reports.
 - Go through the [AIM](#) (8-2) and cover when must a PIREP be submitted.
- G-AIRMETs
 - Graphical advisories for widespread weather hazards that are less severe than SIGMET

- conditions. They cover icing, turbulence, and IFR/Mountain Obscuration hazards.
 - They are issued every 6 hours over the CONUS.
 - They are valid at 3-hour forecast snapshots out to 12 hours into the future, typically at 0, 3, 6, 9, and 12 hours from issuance.
 - The first 6 hours of a G-AIRMET correspond to the old legacy AIRMET forecast period; the product then extends to 12 hours total.
 - G-AIRMET hazard types include:
 - Sierra – IFR conditions and mountain obscuration
 - Tango – turbulence, strong surface winds, and low-level wind shear
 - Zulu – icing and freezing level information
 - On Aviation Weather Center products, G-AIRMETs are shown as specific valid-time snapshots, not one continuous block of time.
- SIGMET (WS) – Advises of severe weather conditions that may be hazardous to aircraft.
 - Unscheduled (comes out when needed).
 - Valid for 4 hours (6 for hurricanes).
 - Severe icing not associated with thunderstorms.
 - Severe or greater turbulence.
 - Widespread Dust/sandstorms.
 - Volcanic ash.
 - Issued with alphabetic identifiers (no sierra or tango) → different zones throughout the USA have designated identifiers.
 - Issued when these weather conditions are forecasted to happen over an area of at least 3000 square miles or an area judged to have a significant impact on safety of aircraft.
- Convective SIGMET (WST) – Hazardous convective weather affecting every flight.
 - Issued by Aviation Weather center
 - Issued at 55 minutes past the hour.
 - Valid for 2 hours.
 - Severe thunderstorms with surface winds greater than 50 knots.
 - Hail at surface at least $\frac{3}{4}$ inch in diameter.
 - Affects 40% or more of a 3000 sq miles area.
 - Line of thunderstorms at least 60 miles long with thunderstorms affecting at least 40 percent of its length.
 - Tornadoes.
 - Numbered sequentially depending on the zone (West, Central, East).
 - If no convective weather, will say "NONE".
- Center Weather Advisory:
 - A CWA is an aviation weather warning for conditions meeting or approaching national in-flight advisory (AIRMET, SIGMET, or Convective SIGMET) criteria.

- CWAs are valid for up to 2 hours and may include forecasts of conditions expected to begin within 2 hours of issuance.
- Issued by ARTCC meteorologists.
- Convective outlook:
 - Consist of both a narrative and a graphic showing severe thunderstorm threats across the US.
 - Convective outlooks are split into 4 different charts:
 - Day 1: risk of severe weather for 24 hours. Issued 5 times a day.
 - Day 2: risk of severe weather for the next 24 hours. Issued 2 times a day.
 - Day 3: risk of severe weather for the next 24 hours. Issued daily.
 - Day 4-8: risk of severe weather for day 4 through day 8. Issued daily.
 - <https://www.weather.gov/jetstream/ahead>
- AWOS, ASOS, and ATIS:
 - Check the chart supplement to see what type of weather system is in use.
 - AWOS:
 - Automated Weather Observing System.
 - Operated and controlled by the FAA.
 - Automated and reports the weather every minute.
 - Types of AWOS:
 - AWOS-A: Altimeter setting.
 - AWOS-AV: Altimeter and visibility.
 - AWOS-I: Altimeter, wind, temperature, dew point, and density altitude.
 - AWOS-II: AWOS-I plus visibility.
 - AWOS-III: AWOS-II plus ceiling.
 - AWOS-III P: AWOS-III plus precipitation identification.
 - AWOS-III T: AWOS-III plus thunderstorm/lightning identification.
 - AWOS-III PT: AWOS-III plus precipitation and thunderstorm/lightning identification.
 - AWOS-IV: AWOS-III plus precipitation, type and accumulation, freezing rain, thunderstorm, and runway surface sensors.
 - ASOS:
 - Automated Surface Observing System.
 - Operated and controlled by the National Weather Service (NWS), Department of Defense (DoD), and sometimes by the FAA.
 - Used for the national weather system, not just for aviation purposes.
 - Reported information:
 - Wind speed and direction.
 - Visibility.
 - Sky condition.
 - Ceiling height and precipitation.

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- Barometric pressure.
 - Density altitude.
- ATIS:
 - Automated Terminal Information Service.
 - Towered airports.
 - Requires a human to monitor and broadcast the ATIS.
 - ATIS information is obtained from the ASOS system.
 - Contains more than just weather (NOTAMs, remarks, etc...)
 - Published hourly at 55 minutes past the hour.
 - If the weather is rapidly changing, the tower can give an updated ATIS (like a SPECI METAR).
 - Reported information:
 - Phonetic alphabet letter.
 - Time of the report.
 - Wind speed and direction.
 - Visibility.
 - Cloud cover.
 - Temperature.
 - Dew point.
 - Barometric pressure.
 - NOTAMs, remarks, other information.
- VFR Weather Minimums
 - Refer to figure 15-8 in the Pilot's Handbook of Aeronautical Knowledge.
- Fog:
 - Temperature of air near the ground is cooled to the air's dew point.
 - Types (reported when visibility is less than 5/8th of a mile).
 - Radiation fog – forms in low-lying areas like mountain valleys. Occurs when the ground cools rapidly due to terrestrial radiation, and surrounding air reaches dew point. As the sun rises and temperature increases, the fog burns off.
 - Advection fog – when a layer of warm moist air moves over a colder surface. A 15-knot wind is necessary. Common to coastal areas where sea breezes can blow the air over cooler landmasses.
 - Upslope fog – Occurs when moist, stable air is forced up sloping land features like a mountain range. Also requires wind for formation and continued existence.
 - Steam fog – or sea smoke, forms when cold dry air moves over warm water. As water evaporates it turns into smoke.
- Visibility and ceilings:
 - Stress that continued flight into deteriorating weather is inadvisable.
 - Discuss the disadvantages of scud running of VFR over the top.
 - Discuss diversion procedures.

- Wind:
 - Refer to hazards posed by exceeding wind limitations.
 - Discuss diversion to a proper alternate with either calmer winds or more headwind.
 - Review the use of V_A and V_{NO} to prevent damage in turbulence.
- Turbulence:
 - Types:
 - Light chop: slight rapid bumpiness.
 - Light turbulence: series of momentary changes in your altitude or attitude.
 - Moderate chop: more intense than light chop.
 - Moderate turbulence: changes in attitude and altitude, positive control though.
 - Severe turbulence: large abrupt changes in altitude and attitude, tempo loss of control.
 - Extreme turbulence: aircraft violently tossed around, impossible to control. Cause structural damage.
 - CAT → severe turbulence in cloudless regions, usually at high altitude with wind shear.
 - Intensities:
 - Light.
 - Moderate.
 - Severe.
 - Extreme.
 - Frequencies:
 - Occasional.
 - Intermittent.
 - Continuous.
- Thunderstorms:
 - Emphasize that thunderstorms possess all known weather hazards.
 - Refer to the Flight Operations Manual for guidance (hold outside/cancel/divert).
 - Thunderstorms:
 - What you need for a thunderstorm:
 - Lifting action.
 - High moisture.
 - Unstable air.
 - Stages:
 - Developing stage
 - Towering cumulus (rising air), little rain, occasional lightning.
 - Mature
 - Fully formed, heavy rain, frequent lightning, strong winds.
 - Dissipating

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- Downdrafts, rainfall decreases, still produce strong winds and lightning.
- Types:
 - Single cell: develops on warm humid days. severe, produce hail, and microburst.
 - Multi cell: develop in clusters, cover large areas, can move in different directions.
 - Squall line: narrow band of TS, ahead of cold front (moist, unstable air, lifting action). Severe and wide.
 - Super cell: Single long-lived TS. large hail and significant tornadoes.
- What to expect:
 - Lightning.
 - Turbulence.
 - Hail.
 - Tornadoes.
 - Wind shear.
- Icing conditions
 - Known icing vs known icing conditions.
 - Known icing: visible accretion.
 - Known icing conditions: Known icing conditions are forecast or reported icing environments. (airmets, freezing level charts, winds and temps aloft, etc...).
 - Structural:
 - Requires visible moisture and aircraft surface below freezing.
 - Clear → forms when large supercooled drop hit the aircraft and spread and freeze slowly
 - Cumulus clouds.
 - Clear in color.
 - Hits leading edge and freezes as it reaches trailing edge.
 - -10 to 0 Celsius.
 - Rime → forms when small supercooled droplets hit the aircraft and freeze rapidly.
 - White color.
 - Stratus clouds.
 - Freezes instantly.
 - -20 to -10 Celsius.
 - Mix → mix of the two.
 - found in thunderstorms where there is a mix of stratus and cumulus clouds.
 - Frost → ice crystal deposits formed by sublimation (solid to gas) when temp and dew point are below freezing.
 - Effects of icing:
 - Drag → increase.
 - lift → decrease.
 - Weight → increase.

- Stall speed → increase.
- Induction:
 - Carburetor ice → Is most likely to occur between 20°F and 70°F (-7°C to 21°C) when relative humidity is high.
 - It can occur even when the outside air temperature is as high as 100°F (38°C) if sufficient moisture is present.
 - Air in the carburetor is accelerated in the venturi, causing the temperature of the air to decrease. More susceptible to icing.
 - Turn carb heat on (small reduction in RPM, warm air is less dense than cold air, enriches the mixture).
- Instrument icing:
 - Pitot tube and static ports.
 - Static wicks.
 - Antennas.
 - Stall warning tab.
 - OAT probe.

National Airspace System

- Controlled airspace (91.127 - 91.135).
 - General term that refers to the different classifications of airspace within which air traffic control (ATC) service is provided.
 - Class A:
 - Generally extends from 18,000 feet MSL (FL180) to FL600 throughout the United States
 - Entry requirements:
 - ATC Clearance.
 - IFR flight plan.
 - Required equipment:
 - Two-way radios.
 - Mode C transponder.
 - ADS B-OUT.
 - Class B:
 - Generally, from SFC to 10,000 ft MSL.
 - Busy airports: based on total operations or passenger enplanements.
 - Multiple layers (upside down cake).
 - Configuration can be different depending on the airport.
 - ATC provides separation within the airspace.
 - Some airports do not allow student, sport, or recreational pilots to land, usually due to high traffic volume
 - Part 91, appendix D, section 04.
 - Entry requirements:
 - ATC Clearance (N640FT, Orlando approach, you are cleared through the Orlando class B).

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- Required equipment:
 - Two-way radios.
 - Mode C transponder (in class B, above and within the lateral limits of class B, and within 30 nm (mode C veil)).
 - ADSB-OUT (in class B, above and within the lateral limits of class B, and within 30 nm (mode C veil)).
- Class C:
 - Generally, from SFC to 4,000 ft MSL.
 - Surrounds airports that have a control tower, radar approach control, and have a certain number of IFR operations or passenger enplanements.
 - Configuration can be different depending on the airport.
 - Two layers:
 - Inner ring extends from the SFC to 4,000 ft (typically) with a 5 NM radius.
 - Outer ring extends from 1,200 ft to 4000 ft (typically) with a 10 NM radius.
 - Entry requirements:
 - Establish two-way radio communication (ATC MUST say your call-sign).
 - Required equipment:
 - Two-way radios.
 - Mode C transponder (in class C and above and within the lateral limits of class C).
 - ADSB-OUT (in class C and above and within the lateral limits of class C).
- Class D:
 - Generally, from SFC to 2,500 ft MSL.
 - Surrounds airports that have a control tower.
 - Configuration can be different depending on the airport (KMLB!).
 - Typically designed to contain instrument approach procedures and transition areas..
 - Entry requirements:
 - Establish two-way radio communication (ATC MUST say your call-sign).
 - Required equipment:
 - Two-way radios.
- Class E:
 - Controlled airspace that is not class A, B, C, or D.
 - Extends up from either the surface or a designated altitude to the overlying or adjacent controlled airspace.
 - Sectional and other charts depict all locations of Class E airspace with bases below 14,500 feet MSL. In areas where charts do not depict a class E base, class E begins at 14,500 feet MSL.

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- In most areas, the Class E airspace base is 1,200 feet AGL. In many other areas, the Class E airspace base is either the surface or 700 feet AGL.
- Class A ends at FL600, and Class E begins above that altitude..
- Required equipment:
 - ADSB-OUT: at or above 10,000 feet msl, excluding airspace at and below 2,500 feet agl; over the Gulf of Mexico, at and above 3,000 feet msl, within 12 nm of the U.S. coast.
 - Mode C transponder: at and above 10,000 feet msl, excluding the airspace at and below 2,500 ft agl.
- Uncontrolled airspace (91.126)
 - Portion of the airspace system that is not designated as A, B, C, D, or E.
 - Class G:
 - Uncontrolled.
 - ATC has no authority or responsibility.
 - Although ATC has no authority, a pilot cannot be careless or reckless as this would break the 91.13 regulation.
 - VFR rules apply.
- Special use airspace (WARMPC):
 - Where certain activities must be confined.
 - Limitations may be imposed on operations not participating.
 - Information for airspaces is located on aeronautical charts.
 - **Warning area:**
 - Warning areas are located over international waters where the U.S. does not have full jurisdiction..
 - Defined dimensions, extending 3NM offshore of the U.S.
 - Contains activity that may be hazardous to nonparticipating aircraft.
 - Charted as "W" followed by a number.
 - **Alert area:**
 - High volume of pilot training or an unusual type of aerial activity.
 - Depicted with an "A" followed by a number on charts.
 - **Restricted area:**
 - Area in which operations are hazardous to non-participating aircraft.
 - Artillery firing, guided missiles.
 - Flight of aircraft may be restricted, but not prohibited.
 - Must have permission from ATC or the controlling agency to enter.
 - Charted with an "R" followed by a number.
 - Information found on the back of charts.
 - **Military Operation Area (MOA):**
 - Defined vertical and lateral limits.
 - Separates certain military training activities from IFR traffic.
 - IFR traffic may be cleared through airspace if ATC can provide separation.

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- Additional information for each found on the back of the sectional.
- VFR aircraft may fly through MOAs without clearance, but should exercise caution
- **Prohibited area:**
 - Airspace in which flight is prohibited.
 - Camp David, White House, Congressional buildings.
 - Usually for security or national welfare.
 - Charted at "P" followed by a number.
- **Controlled firing area:**
 - Contains activity that is hazardous to non-participating aircraft.
 - Activity must stop if spotter spots non-participating aircraft.
 - Published in chart supplement under Special Notices.
- **Other airspace:**
 - **Special Air Traffic Rules (SATR) and Special Flight Rules Area. (SFRA)**
 - An area designated by the FAA to have additional rules and regulations which must be followed while operating inside of.
 - Some examples of SFRAs are Washington DC and The Grand Canyon.
 - There may be additional training required, or you may be required to communicate with ATC in situations where normally you would not need to.
 - It is critically important that a pilot makes themselves aware of all rules that might apply inside of a SFRA before making a flight inside of one.
 - **Military Training Routes (MTRs):**
 - Used by military aircraft to maintain proficiency in tactical flying.
 - Usually established below 10,000 feet MSL for operations exceeding 250 knots.
 - Routes depicted as IR (if IFR) and VR (if VFR), followed by a number:
 - Those not exceeding 1,500 feet AGL contain 4 numbers (IR 1206).
 - Those above 1,500 feet AGL contain 3 numbers (VR 207).
 - **Parachute jump aircraft operations:**
 - Airports at which there are high numbers of parachute operations.
 - Published in chart supplement under Parachute Jumping Areas and depicted on sectional charts.
 - **Published VFR Routes:**
 - Used for transitioning around, under, or through complex airspace.
 - Usually found on VFR terminal area planning charts.
 - Often referred to as VFR flyway, VFR corridor, Class B airspace VFR transition route, and terminal area VFR route.
 - **Terminal Radar Service Areas (TRSAs):**
 - Areas in which pilots can receive additional radar services.
 - Purpose is to provide separation for IFR and participating VFR traffic.
 - The airspace within the TRSA becomes Class D around the airports.

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- Remaining portion overlies other controlled airspace, which is normally Class E.
- Depicted on VFR sectional charts and terminal area charts with a solid black line and altitudes for each segment.
- Participation is voluntary and encouraged.
- Temporary flight restrictions (TFRs):
 - Designated by flight data center (FDC) Notice to Airmen (NOTAM):
 - Begins with phrase “FLIGHT RESTRICTIONS” followed by location, effective time period, and defined in statute miles, and altitudes affected.
 - Purposes may include:
 - Protect persons or property in air or on surface from existing or imminent hazard.
 - Provide a safe environment for operation of disaster relief aircraft.
 - Prevent unsafe congestion of sightseeing aircraft above incident or event.
 - Protect declared national disasters for humanitarian reasons in Hawaii.
 - Protect the President, Vice President, or other public figures.
 - Provide a safe environment for space agency operations.
 - If penetrated, the pilot may undergo security investigations and certificate suspensions.
- National Security Areas
 - National Security Areas (NSAs) are designated areas where increased security and safety of ground facilities are required. These areas are established by the FAA in cooperation with federal security agencies.
 - NSAs are typically located over sensitive facilities, such as military installations or other areas of national importance.
 - Pilots are requested to voluntarily avoid flying through these areas.

Cross-Country Flight Planning

- Flight planning and navigation log:
 - Refer to the VFR cross-country flight planning guide on the Florida Tech Aviation hub or the Training Drive.
- NOTAMs, AIM, Chart Supplement:
 - Notice to Air Missions (NOTAMs):
 - Contains pertinent information about aircraft hazards that can affect a flight.
 - This information is not known far enough in advance to be publicized by another means.
 - For example: a taxiway diagram will not receive an immediate change if there is a taxiway closure. A NOTAM will be issued instead.
 - Types ([AIM 5-1-3](#)):
 - NOTAM (D).

- FDC NOTAM.
 - E.g. TFRs
 - International NOTAM.
 - Military NOTAM.
 - Decoding a NOTAM:
 - AIM 5-1-3.
 - Ensure the student is familiar with the various sections of the [AIM](#).
 - Chapter 01 - Air navigation.
 - Chapter 02- Aeronautical lighting.
 - Chapter 03 - Airspace.
 - Chapter 04 - Air traffic control.
 - Chapter 05 - Air traffic procedures.
 - Chapter 06 - Emergency procedures.
 - Chapter 07 - Safety of flight
 - Chapter 08 - Medical facts.
 - Chapter 09 - Aeronautical charts.
 - Chart supplement:
 - Open the chart supplement to Melbourne or another airport and go through all of the information with the student.
 - If they don't, see if they can reference the legend.
 - Students should identify:
 - Airport diagram
 - Runway lengths
 - Lighting
 - Pattern altitude
 - CTAF frequency
 - Fuel availability
 - Remarks
- Alternate planning:
 - It is always a good idea to plan for an alternate in the event of a malfunction, deteriorating weather, aeromedical factors, sudden airport closure, etc.
 - Discuss how to pick an appropriate alternate airport.
- Pilotage and dead reckoning:
 - Pilotage: navigation by reference to landmarks or checkpoints.
 - A method of navigation that can be used on any course with adequate checkpoints but is more commonly used with dead reckoning and VHF radio navigation.
 - The checkpoint used should be prominent features common to the area of flight.
 - Roads, rivers, railroad tracks, lakes, and power lines.
 - If possible, select features that will make useful boundaries on each side of the course.
 - Keep from drifting too far off course by referring to and not crossing

selected points.

- Never place complete reliance on any single checkpoint, choose ample checkpoints.
 - If the checkpoint is not in sight, do not continue blindly.
- Dead reckoning: navigation solely by computation based on time, airspeed, distance, and direction.
 - The products derived from these, when adjusted by wind speed and velocity, are heading and ground speed (GS).
 - The predicted heading will guide the airplane along the intended path and the GS will establish the time to arrive at each checkpoint and destination.
- ATC communications/ Flight following:
 - Flight Service Station (FSS):
 - This is who needs to be called to open the flight plan.
 - Look at the sectional chart to see which frequency to use and who to call.
 - Opening the flight plan:
 - Pilot: "St-Petersburg radio, N640FT, transmitting and receiving on 122.2".
 - FSS: "N640FT, St. Petersburg radio, go ahead".
 - Pilot: "St-Petersburg radio, N640FT would like to open our flight plan from KMLB to KSUA".
 - FSS: "N640FT, St. Petersburg radio, flight plan is activated".
 - Flight plans can also be opened automatically through 1800wxbrief.com or EFB applications such as ForeFlight.
 - Closing the flight plan:
 - Pilot: "St-Petersburg radio, N640FT, transmitting and receiving on 122.2"
 - FSS: "N640FT, St. Petersburg radio, go ahead".
 - Pilot: "St-Petersburg radio, N640FT would like to close our flight plan from KMLB to KSUA".
 - FSS: "N640FT, St. Petersburg radio, flight plan is terminated".
 - Flight following:
 - Flight following is extremely useful when it comes to cross-country flights.
 - Traffic advisories.
 - Airport information.
 - Weather updates.
 - Requesting flight following:
 - Use the chart supplement to obtain the desired frequency.
 - Pilot: "Orlando approach, Tech Flight 615, 6 miles south west of Melbourne at 2000ft, type Piper Cherokee, requesting flight following from KMLB to KSUA,".
 - ATC: "Tech Flight 615, Orlando approach, squawk 1234".
 - Pilot: "squawk 1234 , Tech Flight 615".
 - ATC: "Tech Flight 615, Orlando approach, radar contact, altimeter is 30.01".
 - Enroute:

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- ATC will give you traffic advisories, vectors, frequency changes, etc....
- Actively listen, read back the instructions, and comply!
- Arriving:
 - It may be necessary to request a temporary frequency change to close the flight plan with FSS.
 - Pilot: "Orlando approach, N640FT, requesting frequency change to close flight plan".
 - ATC: "N640FT, Orlando approach, approved and check back in when on frequency".
 - Pilot: "Orlando approach, N640FT, is back with you".
 - Hand-off:
 - ATC: "N640FT, Orlando approach, contact KSUA tower on 126.6"
 - Pilot: "contact tower on 126.6, N640FT."
 - Canceling flight following:
 - Pilot: "Orlando approach, N640FT would like to cancel flight-following".
 - ATC: "N640FT, Orlando approach, radar services terminated, squawk VFR, frequency change approved".
 - Pilot: "Squawk VFR, frequency change approved, N640FT".
- Navigation systems:
 - GPS:
 - Before taxiing, enter the route of flight into the GPS.
 - The GPS will provide information similar to the Nav Log and update distances, time enroute, fuel burn, etc. as changes are made to the flight plan.
 - This saves you tons of work and time, especially in the case of a diversion or emergency.
 - The GPS is an extremely useful tool and should be used once the pilot is capable of competently navigating with it.
 - Great for situational awareness.
 - Ensure you don't get complacent with the GPS and lose situational awareness, a GPS failure in this case could be hazardous.
 - If the GPS fails, you must be able to safely continue along the route by means of pilotage and dead reckoning.
 - VOR:
 - Use the sectional chart to figure out which radial to track.
 - Make sure the VOR frequency and radials are written on the navlog.
 - The student must tune and ID the VOR prior to using it.
 - Show on the sectional chart how to use a VOR airway/radial to get from point to point.

Diversion

- There may be a time where the flight cannot be continued to the planned destination.

- Weather, malfunctions, poor planning, fuel, pilot/passenger, fatigue/illness, etc.
- Before flight, check the route for suitable landing areas and for nav aids that can be used in a diversion.
- Take advantage of all shortcuts/rule of thumb computations when computing course/speed/distance.
 - Use your thumb to estimate distance:
 - Using the scale on your map, figure out approximately how far from the fingertip of your thumb (toward the knuckle) 10nm is.
 - This will provide a rough distance estimate for quicker, less stressful time and fuel calculations.
 - Use a compass rose, airway or any other reference to determine the approximate new heading.
- Procedure:
 - Confirm your present position on the sectional chart.
 - Choose an alternate shown on your sectional or use the 'Nearest' page on your GPS.
 - Divert immediately toward the alternate using shortcuts/rule of thumb calculations (above) if it is an emergency. If time permits, circle until all of the information is obtained (heading, time, frequencies, etc...).
 - Fill out the diversion log.

Regulations Relevant to Cross-Country Flight

- 61.87 – Required pre-solo endorsements
 - Pre-solo aeronautical knowledge test - 61.87(b).
 - 61.87(c) – Pre-solo flight training
 - 61.87(n) – Solo flight (initial 90 days)
- 61.93 – Require solo cross-country endorsements.
 - 61.93(c)1 and (c)2 – Cross-country flight training.
 - 61.93(c)3 – Solo cross-country flight (specific route).
- 91.151 – Fuel Requirements for VFR Flight
 - No person may begin a flight under VFR conditions unless sufficient fuel is available to fly:
 - Day VFR: to the first point of intended landing plus 30 minutes at normal cruising speed.
 - Night VFR: to the first point of intended landing plus 45 minutes at normal cruising speed.
 - Review fuel minimums per FOM
- 91.159 – VFR Cruising Altitudes
 - When operating more than 3,000 feet above the surface, pilots must maintain appropriate cruising altitudes based on magnetic course:
 - Magnetic course 0°–179°: odd thousands + 500 feet (e.g., 3,500 ft).
 - Magnetic course 180°–359°: even thousands + 500 feet (e.g., 4,500 ft).
- 91.126–91.135 – Operations in Different Classes of Airspace
 - These sections describe operating requirements within Class G, E, D, C, B, and A

airspace, including communication and equipment requirements.

Operations of Systems

- Review lesson 109

Systems and Equipment Malfunctions

- During cross-country flight planning, pilots must be prepared to recognize and respond to potential aircraft system or equipment malfunctions. Understanding common failures and appropriate corrective actions helps ensure safe flight operations.
- Pilots should be familiar with:
 - Electrical system failures, such as alternator or battery malfunctions.
 - Engine-related issues, including rough running engines or abnormal indications.
 - Navigation equipment failures, such as loss of GPS or VOR signals.
 - Instrument failures, including malfunctioning flight or engine instruments.
- If a malfunction occurs, the pilot should:
 - Maintain aircraft control and analyze the situation.
 - Refer to the appropriate checklists in the POH if time permits.
 - Determine whether the flight can be safely continued or if a diversion to the nearest suitable airport is required.
 - Communicate with ATC or Flight Service if assistance is needed.
- Proper preflight inspection, monitoring of aircraft systems during flight, and familiarity with emergency procedures help reduce the risks associated with system and equipment malfunctions.

Emergency Equipment and Survival Gear

- When planning a cross-country flight, pilots should consider the availability of emergency equipment and survival gear, particularly when flying over remote areas or terrain where assistance may not be immediately available.
 - Common emergency equipment may include:
 - First aid kit
 - Flashlight with extra batteries
 - Emergency locator transmitter (ELT)
 - Fire extinguisher
 - Portable radio or backup communication device
 - Water and basic survival supplies
- Pilots should also consider the environment and terrain along the route of flight. For example, flights over mountains, forests, or large bodies of water may require additional survival equipment.
- Proper preparation helps ensure that pilots and passengers are equipped to manage an emergency situation until help arrives.

Lost Procedures

- 7C's:

- Calm
- Climb
 - Identify prominent landmarks.
- Circle.
 - Avoid aggravating the situation.
- Crosscheck
 - GPS.
 - VOR: use 2 VOR's, center the CDI, draw on the sectional chart to determine position.
- Communicate
 - Request assistance from ATC.
- Comply
 - Follow ATC instructions.
- Conserve
 - reduce power to 55% and lean the mixture for best economy (refer to PIM).

Land and Hold Short Operations (LAHSO)

- Review [Land and Hold Short Operations](#) (LAHSO):
 - Who can accept it?
 - Do you have to accept it?
 - When can you deviate after accepting?
 - Show an example on an airport diagram (KSRQ, KPBI, KTPA).
 - AIM 4-3-11 - Pilot Responsibilities When Conducting LAHSO.

Controlled Flight into Terrain (CFIT)

- Review Controlled Flight Into Terrain (CFIT):
 - Unintentional collision with terrain (ground, mountain, water, or obstacle).
 - Commonly occurs on approach or during the landing phase of a flight.
 - How does it happen?
 - Loss of situational awareness
 - Procedural mistakes (descent below MDA while IFR).
 - Aircraft performance (high density altitude, tailwind approaches, etc...).

Wire Strike Avoidance

- Review from lesson 116

Aviation Security

- Review from lesson 116

Single-Pilot Resource Management (SRM)

- Review from lesson 116

Completion Standards

This lesson is complete when the applicant can:

1. Understand the concepts behind the maneuvers introduced.
2. Understand how to perform the maneuvers introduced.